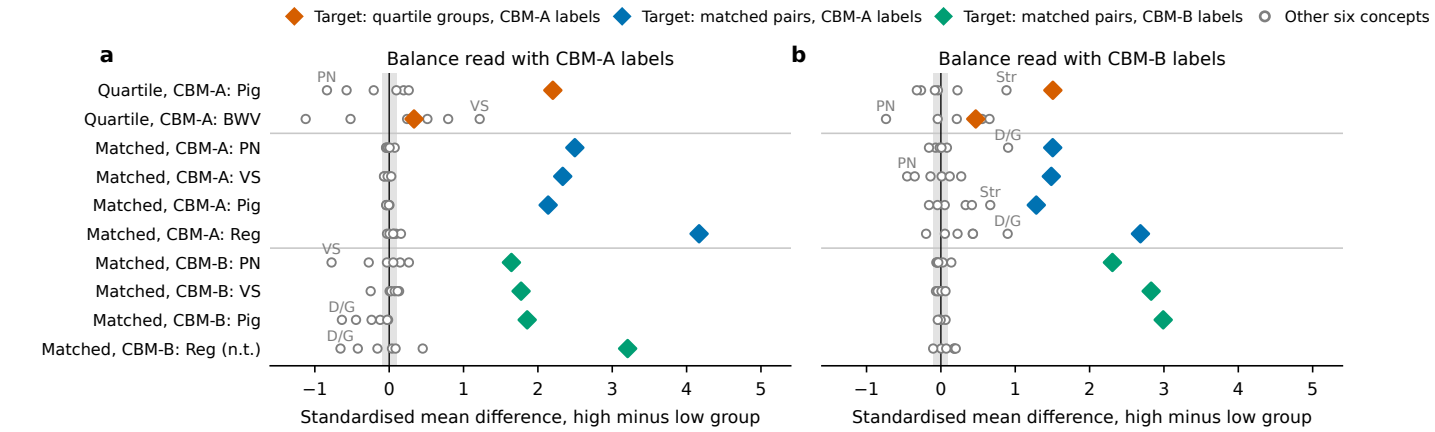
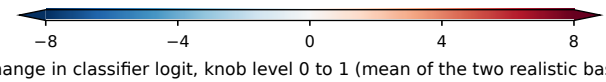


c

Original and CBM-A matched controls

		PN knob	BWV knob	VS knob	Pig knob	Str knob	D/G knob	Reg knob	Bases passed
Original	Quartile Pig	-3.6	+1.7	-1.3	+8.3	+3.3	-0.3	-0.7	4/4
	Quartile BWV	-3.2	+1.2	+4.9	+4.9	-0.7	+0.3	+4.1	0/4
	Permuted labels	+0.1	+0.0	+0.0	+0.0	+0.0	+0.0	+0.1	neg.
Matched PN	ResNet-50 s1	+4.1	-0.2	-2.8	-3.8	-2.0	+0.5	-0.9	4/4
	ResNet-50 s2	+4.5	+0.0	-2.9	-3.6	-2.0	+0.6	-0.4	4/4
	ResNet-50 s3	+4.3	-0.4	-2.8	-4.1	-2.2	+0.4	-0.6	4/4
	ViT-B/16 s1	+6.4	+0.2	-3.8	-5.0	-2.3	+0.4	-0.2	4/4
Matched VS	ResNet-50 s1	-1.3	-0.7	+6.0	-5.8	-3.9	-0.1	+1.8	4/4
	ResNet-50 s2	-1.8	-0.5	+6.1	-3.5	-3.6	-0.2	+1.9	4/4
	ResNet-50 s3	-1.9	-0.0	+6.3	-3.4	-3.3	-0.2	+2.0	4/4
	ViT-B/16 s1	-0.8	-0.1	+4.9	-2.3	-1.8	-0.4	+1.3	4/4
Matched Pig	ResNet-50 s1	-0.9	+1.6	-1.3	+5.8	+1.5	+0.1	+0.1	4/4
	ResNet-50 s2	-2.7	+1.2	-2.0	+8.1	+2.6	-0.3	-1.5	4/4
	ResNet-50 s3	-1.9	+1.5	-2.4	+8.8	+2.6	-0.2	-1.7	4/4
	ViT-B/16 s1	-1.6	+1.8	-1.6	+5.9	+2.1	+0.1	-0.2	4/4
Matched Reg	ResNet-50 s1	+1.0	+1.4	+4.2	-1.2	-3.2	+0.1	+4.2	3/4
	ResNet-50 s2	+1.2	+1.2	+3.1	-1.5	-2.4	-0.1	+3.3	4/4
	ResNet-50 s3	+1.2	+1.5	+3.6	-1.1	-2.8	+0.1	+3.8	3/4
	ViT-B/16 s1	+1.5	+0.5	+3.0	-1.6	-1.8	-0.2	+3.1	3/4

d

CBM-B matched controls

	PN knob	BWV knob	VS knob	Pig knob	Str knob	D/G knob	Reg knob	Bases passed	
PN	ResNet-50 s1	+4.5	+0.6	-1.9	-2.2	-1.5	-0.1	-0.2	4/4	
	ResNet-50 s2	+4.7	+0.3	-2.4	-3.1	-1.3	+0.0	-0.7	4/4	
	ResNet-50 s3	+4.4	-0.2	-2.4	-3.0	-1.2	-0.3	-0.6	4/4	valid
	ViT-B/16 s1	+6.3	+1.1	-4.2	-2.6	-1.5	-0.1	-0.5	4/4	
VS	ResNet-50 s1	-1.3	-0.7	+5.1	-5.5	-3.4	-0.5	+1.4	4/4	
	ResNet-50 s2	-0.8	-0.1	+4.4	-3.2	-2.8	-0.3	+1.3	4/4	
	ResNet-50 s3	-0.8	-0.2	+4.6	-3.2	-3.2	-0.5	+1.6	4/4	valid
	ViT-B/16 s1	-2.0	+0.1	+8.5	-3.5	-4.0	-0.6	+3.9	4/4	
Pig	ResNet-50 s1	-0.1	+0.1	-0.1	+0.3	+0.1	-0.0	-0.1	4/4	
	ResNet-50 s2	-0.1	+0.2	-0.3	+0.9	+0.4	-0.0	-0.1	4/4	
	ResNet-50 s3	-0.0	+0.1	-0.1	+0.3	+0.1	-0.0	-0.0	4/4	valid
	ViT-B/16 s1	-0.6	+0.7	-0.7	+3.3	+1.1	-0.3	+0.0	4/4	
Reg	ResNet-50 s1	not testable: too few pairs							n.t.	
	ResNet-50 s2	not testable: too few pairs							n.t.	
	ResNet-50 s3	not testable: too few pairs							n.t.	not valid
	ViT-B/16 s1	not testable: too few pairs							n.t.